Confounders, colliders, and mediators (answer key)

Each of the following case studies asks how a predictor x affects a response y. Each one features a third variable that is a **confounder**, a **collider**, or a **mediator** (one of each, but you have to figure out which is which). For each case study:

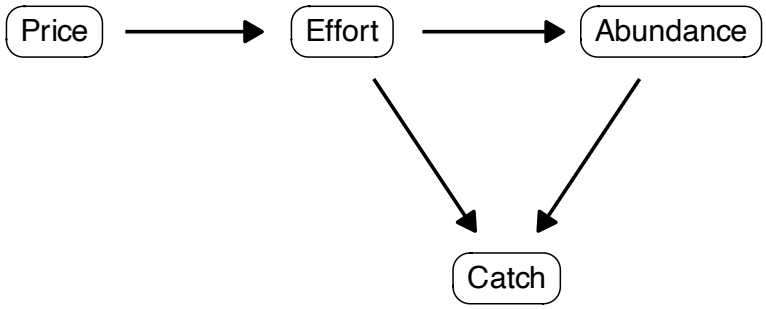
1. **Paths.** Use the DAG to write out every path from x to y, including the direction of each arrow.
2. **Backdoors.** Identify the backdoor paths. Is each one open or closed?
3. **Adjustment.** Decide which variables to include in the model and which to exclude. Is the third variable a confounder, collider, or mediator?
4. **Estimates.** Two researchers fit two different models to the same data. Use the `summary()` and `confint()` output to draw each model's coefficient estimate for x, with its 95% confidence interval, on the blank axes. Then draw a vertical line for the true effect of x on y in a plausible position. If the third variable is a mediator, there's no single true effect. Instead, draw one line for the **total effect** and one for the **direct effect**.
5. **Verdict.** Which model's estimate is biased, and which is unbiased? If the third variable is a mediator, neither model is biased. Instead, decide which model estimates the total effect and which estimates the direct effect.

Remember the backdoor criterion. A set of variables satisfies it if adjusting for them (1) closes every backdoor path between x and y, and (2) doesn't include any variables caused by x.

1. Fishing effort and rockfish abundance

How much does fishing pressure deplete a rockfish population? A state fisheries agency studied 50 fishing grounds along the coast. Divers surveyed each ground to estimate rockfish abundance, independently of the fishery. Logbooks recorded the fishing **effort** in each ground (hundreds of boat-days per year) and the total **catch** (tonnes per year). Catch depends on both how hard boats fish and how many fish there are to catch. Effort is higher where the market **price** of rockfish is higher.

Question: how does fishing effort (`effort`) affect rockfish abundance (`abundance`)?



Paths from Effort to Abundance

1. Effort → Abundance (causal)
2. Effort → Catch ← Abundance

Price isn't on any path between Effort and Abundance. Its only arrow points into Effort.

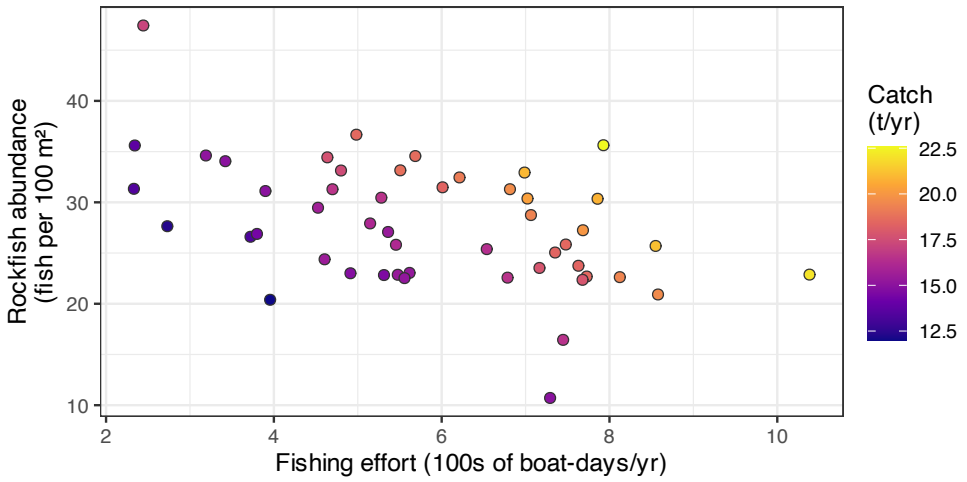
Backdoor paths

There are **no** backdoor paths (no path starts with an arrow into Effort). Path 2 isn't causal, but it's **closed**, because Catch is a collider.

Adjustment

Catch is a **collider**. Exclude it so path 2 stays closed. Price doesn't need to be included.

The data



Two models

$$\textbf{Model A:} \quad \text{abundance} = \beta_0 + \beta_1 \cdot \text{effort} + \varepsilon$$

$$\textbf{Model B:} \quad \text{abundance} = \beta_0 + \beta_1 \cdot \text{effort} + \beta_2 \cdot \text{catch} + \varepsilon$$

Model A

```
fish_a <- lm(abundance ~ effort, data = fish)
summary(fish_a)
```

```

Call:
lm(formula = abundance ~ effort, data = fish)

Residuals:
    Min      1Q  Median      3Q     Max 
-14.839  -3.482  -0.365   3.536  14.456 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  36.7211     2.5785   14.24  < 2e-16 ***
effort       -1.5316     0.4219   -3.63 0.000686 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.435 on 48 degrees of freedom
Multiple R-squared:  0.2154,    Adjusted R-squared:  0.1991 
F-statistic: 13.18 on 1 and 48 DF,  p-value: 0.0006858
```

```
confint(fish_a)
```

```
                2.5 %     97.5 %
(Intercept) 31.536625 41.9056675
effort      -2.379875 -0.6833627
```

Model B

```
fish_b <- lm(abundance ~ effort + catch, data = fish)
summary(fish_b)
```

```

Call:
lm(formula = abundance ~ effort + catch, data = fish)

Residuals:
    Min      1Q  Median      3Q     Max 
-4.2282 -1.0738  0.0632  0.8329  3.6773 

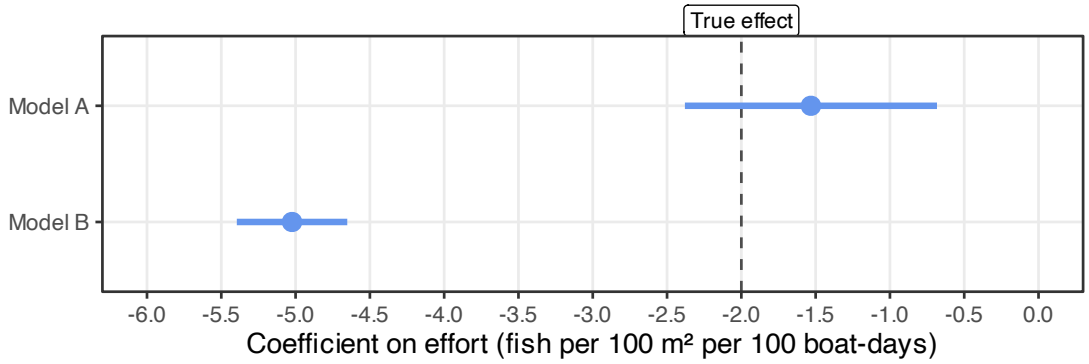
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.5917     1.6484   0.359    0.721    
effort       -5.0238     0.1847 -27.194   <2e-16 ***
catch         3.3020     0.1360  24.273   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.493 on 47 degrees of freedom
Multiple R-squared:  0.942, Adjusted R-squared:  0.9396 
F-statistic: 381.9 on 2 and 47 DF,  p-value: < 2.2e-16
```

```
confint(fish_b)
```

```
                2.5 %    97.5 %
(Intercept) -2.724431  3.907877
effort      -5.395437 -4.652144
catch        3.028315  3.575654
```

Estimates



Verdict

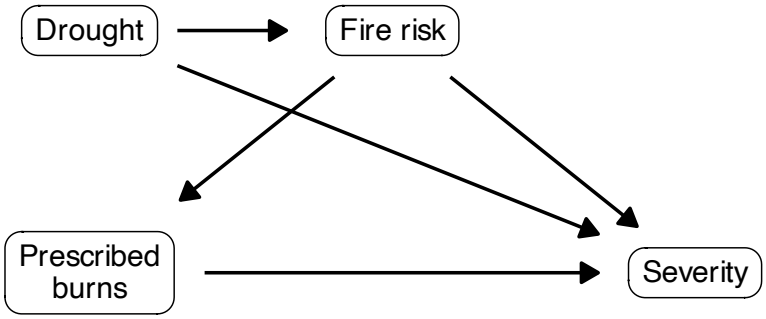
Model A is unbiased. With no backdoors, the simple model recovers the true effect (the simulation used −2).

Model B is biased. Adjusting for catch opens path 2. Among grounds with the *same* catch, the ones that needed more effort to land it must have fewer fish, so Model B exaggerates how much fishing depletes rockfish. This is why catch is a risky stand-in for abundance in fisheries data.

2. Prescribed burns and wildfire severity

Prescribed burns remove dead wood and brush, so they should make later wildfires less severe. A national forest compared 60 management units that burned in wildfires. For each unit, they recorded the percent of the unit treated with prescribed burns in the decade before the wildfire, and the percent that burned at high severity. Managers don't choose units at random. They prioritize units with a high **fire risk** rating, which is based on slope, fuel, and dryness, and is assessed before treatment. Risky units also burn more severely. Regional **drought** raises fire risk, and it also makes wildfires more severe directly.

Question: how does prescribed burning (`burned`) affect wildfire severity (`severity`)?



Paths from Prescribed burns to Severity

1. Prescribed burns → Severity (causal)
2. Prescribed burns ← Fire risk → Severity
3. Prescribed burns ← Fire risk ← Drought → Severity

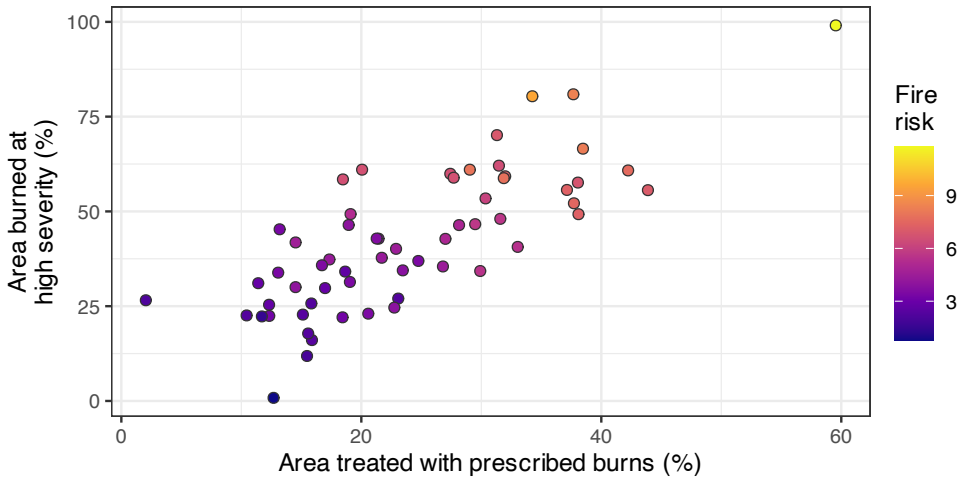
Backdoor paths

Paths 2 and 3 both start with an arrow into Prescribed burns, and neither passes through a collider. Both are **open** backdoors.

Adjustment

Fire risk is a **confounder**. Including it closes both backdoors, because both pass through Fire risk. Drought doesn't need to be included (Fire risk already blocks path 3), though including it wouldn't add bias.

The data



Two models

$$\textbf{Model A:} \quad \text{severity} = \beta_0 + \beta_1 \cdot \text{burned} + \varepsilon$$

$$\textbf{Model B:} \quad \text{severity} = \beta_0 + \beta_1 \cdot \text{burned} + \beta_2 \cdot \text{risk} + \varepsilon$$

Model A

```
fire_a <- lm(severity ~ burned, data = fires)
summary(fire_a)
```

```

Call:
lm(formula = severity ~ burned, data = fires)

Residuals:
    Min      1Q  Median      3Q     Max 
-25.823  -7.147  -1.165   6.885  24.149 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   9.0639     3.8339   2.364   0.0214 *  
burned        1.3855     0.1464   9.466 2.27e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.64 on 58 degrees of freedom
Multiple R-squared:  0.6071,    Adjusted R-squared:  0.6003 
F-statistic: 89.61 on 1 and 58 DF,  p-value: 2.275e-13
```

```
confint(fire_a)
```

```
               2.5 %    97.5 %
(Intercept) 1.389432 16.738322
burned      1.092557  1.678511
```

Model B

```
fire_b <- lm(severity ~ burned + risk, data = fires)
summary(fire_b)
```

```

Call:
lm(formula = severity ~ burned + risk, data = fires)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.0383  -4.0654  -0.3194   2.9088  12.9822 

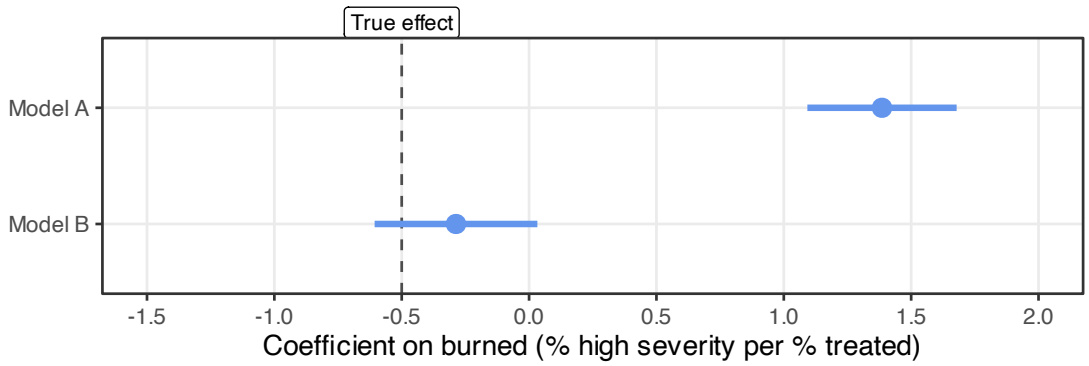
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   6.4121     2.0664   3.103  0.00298 ** 
burned       -0.2869     0.1595  -1.798  0.07740 .  
risk          8.8685     0.7366  12.040  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.237 on 57 degrees of freedom
Multiple R-squared:  0.8891,    Adjusted R-squared:  0.8852 
F-statistic: 228.5 on 2 and 57 DF,  p-value: < 2.2e-16
```

```
confint(fire_b)
```

```
                2.5 %      97.5 %
(Intercept)  2.2742269 10.55003790
burned      -0.6063475  0.03254533
risk         7.3934564 10.34353336
```

Estimates



Verdict

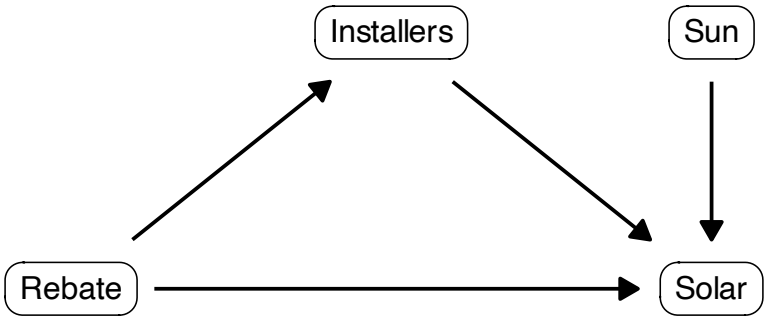
Model A is biased. It omits the confounder, so the open backdoors make prescribed burns look *harmful*. The sign is wrong. Units that were burned were the risky ones, and risky units burn more severely.

Model B is unbiased. Comparing units with the same fire risk, more prescribed burning means less severe wildfire (the simulation's true effect is -0.5).

3. Solar rebates and rooftop solar

Utility rebates make rooftop solar cheaper, which should convince more households to install it. Rebates also attract solar **installers**, who recruit customers door to door. An energy agency compared 60 counties served by different utilities, recording each utility's **rebate** (hundreds of dollars per kW), installers per 100,000 residents, and the percent of homes with rooftop **solar**. Sunnier counties (more peak **sun** hours) also have more rooftop solar.

Question: how do rebates (`rebate`) affect rooftop solar adoption (`solar`)?



Paths from Rebate to Solar

1. Rebate → Solar (causal)
2. Rebate → Installers → Solar (causal)

Sun isn't on any path between Rebate and Solar. Its only arrow points into Solar.

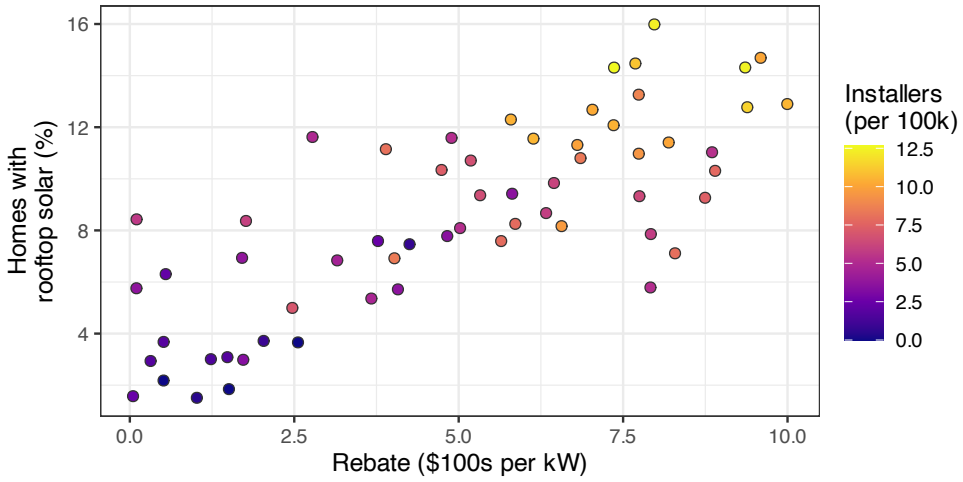
Backdoor paths

There are **no** backdoor paths. Both paths are causal: every arrow points toward Solar.

Adjustment

Installers is a **mediator**. It depends on the question. Exclude Installers for the **total effect** of rebates. Include Installers for the **direct effect** (the rebate's effect on adoption *not* through installers). Sun doesn't need to be included.

The data



Two models

$$\textbf{Model A:} \quad \text{solar} = \beta_0 + \beta_1 \cdot \text{rebate} + \beta_2 \cdot \text{installers} + \varepsilon$$

$$\textbf{Model B:} \quad \text{solar} = \beta_0 + \beta_1 \cdot \text{rebate} + \varepsilon$$

Model A

```
solar_a <- lm(solar ~ rebate + installers, data = counties)
summary(solar_a)
```

```

Call:
lm(formula = solar ~ rebate + installers, data = counties)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.7867 -1.0694 -0.1382  1.1765  4.7433 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   2.5400     0.4923   5.159 3.26e-06 ***
rebate        0.3506     0.1254   2.796  0.00704 ** 
installers    0.6820     0.1046   6.521 1.99e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.82 on 57 degrees of freedom
Multiple R-squared:  0.7707,    Adjusted R-squared:  0.7627 
F-statistic: 95.79 on 2 and 57 DF,  p-value: < 2.2e-16
```

```
confint(solar_a)
```

```
                2.5 %    97.5 %
(Intercept) 1.55416472 3.5258051
rebate      0.09949745 0.6016701
installers  0.47256327 0.8914050
```

Model B

```
solar_b <- lm(solar ~ rebate, data = counties)
summary(solar_b)
```

```

Call:
lm(formula = solar ~ rebate, data = counties)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.5774 -1.8753 -0.2718  1.9544  5.2871 

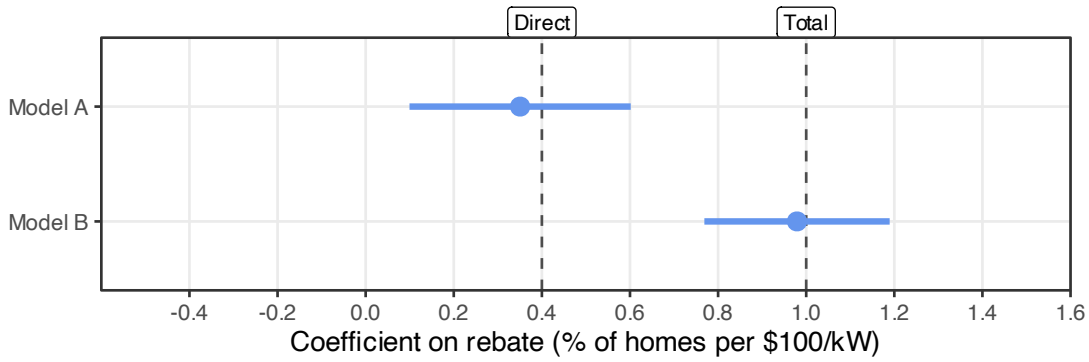
Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   3.6147     0.6077   5.948 1.66e-07 ***
rebate        0.9791     0.1051   9.320 3.94e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.384 on 58 degrees of freedom
Multiple R-squared:  0.5996,    Adjusted R-squared:  0.5927 
F-statistic: 86.87 on 1 and 58 DF,  p-value: 3.943e-13
```

```
confint(solar_b)
```

```
               2.5 %   97.5 %
(Intercept) 2.398281 4.831085
rebate      0.768850 1.189426
```

Estimates



Verdict

Neither model is biased. They answer different questions.

Model A estimates the direct effect. Holding the number of installers constant, each extra $100/kW of rebate raises adoption by about 0.4 percentage points, because solar is cheaper.

Model B estimates the total effect. Each extra $100/kW raises adoption by about 1 percentage point: 0.4 directly, plus $0.8 \times 0.75 = 0.6$ through installers.